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REVIEW

Suicidal ideation during the perinatal period: An umbrella review of prevalence and risk factors



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Abstract The perinatal period involves significant psychological changes that increase women's vulnerability to mental health disorders, including suicidal ideation. This umbrella review, registered in PROSPERO (CRD42024596880), aimed to synthesize evidence on the prevalence of suicidal ideation during this period and its associated risk factors. A search was conducted up to November 2024. Systematic reviews focusing on suicidal ideation and attempts during the perinatal period were included. Methodological quality was assessed using the AMSTAR-2 tool, and overlap was measured using the Corrected Covered Area (8.31%). Ten reviews were analyzed. Prevalence of suicidal ideation ranged from 7.2% to 12%, and suicide attempts from 4% to 5.1%. Common risk factors included poor sleep quality, intimate partner violence, low socioeconomic status, prior mental health disorders, and limited social support. Standardized screening and targeted interventions are important for improving maternal mental health outcomes.

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PALABRAS CLAVE

Revisión paraguas;
Período perinatal;
Postparto;
Embarazo;
Ideación suicida;
Riesgo suicida

Ideación suicida durante el período perinatal: una revisión paraguas sobre la prevalencia y los factores de riesgo

Resumen El periodo perinatal conlleva importantes cambios psicológicos que aumentan la vulnerabilidad de las mujeres a trastornos mentales, incluida la ideación suicida. Esta revisión tipo paraguas registrada en PROSPERO (CRD42024596880), tuvo como objetivo sintetizar la evidencia sobre la prevalencia de la ideación suicida en este periodo y sus factores de riesgo. Se realizó una hasta 2024. Se incluyeron revisiones sistemáticas sobre ideación y tentativa suicidas en este periodo. La calidad metodológica se evaluó mediante AMSTAR-2 y la superposición con el Corrected Covered Area (8.31%). Se analizaron 10 revisiones. La prevalencia de ideación suicida osciló entre 7,2% y 12% y los intentos de suicidio entre 4% y 5,1%. Los factores de riesgo más frecuentes fueron: mala calidad del sueño, violencia de pareja, bajo nivel socioeconómico, antecedentes de trastornos mentales y escaso apoyo social. Se recomienda activar intervenciones específicas estandarizadas.

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Introduction

The perinatal period, defined as the time spanning from pregnancy through the first year after birth, is characterized by profound physiological and psychosocial changes.¹ During this period, mental disorders are common and represent one of the most frequent complications of pregnancy, associated with increased maternal and fetal morbidity and mortality.² The World Health Organization (WHO) estimates that globally, between 10% and 16% of pregnant women, and between 13% and 20% of postpartum women, experience some form of mental disorder.³ As there has been growing attention given to perinatal mental health issues, research has traditionally focused on postpartum depression and puerperal psychosis, leaving other conditions, such as suicidal ideation, relatively unexplored.⁴ Suicidal ideation refers to thoughts about suicide and is distinct from self-harm or non-suicidal self-injury, which aim to alleviate suffering rather than to end one's life.^{5,6} Given its strong association with subsequent suicide attempts and completions, evaluating suicidal ideation during the perinatal period should be a clinical priority. Utilizing validated screening tools offers an opportunity for timely interventions that can prevent more severe suicidal behaviors,⁷ as the presence of suicidal ideation not only increases the risk of suicide and self-harm but is also linked to adverse obstetric outcomes, such as miscarriage, preterm birth, and fetal death.^{8,9}

Contrary to the belief that pregnancy protects against suicide,^{10,11} evidence suggests that suicidal ideation may be more prevalent in pregnant women than in the general population. Studies report a prevalence of suicidal thoughts during pregnancy in 33% of women, compared to 5–15% postpartum.⁹ Given the high comorbidity with depression and potential consequences, universal screening and preventive interventions are urgently needed to support at-risk pregnant women.³ Identified risk factors for suicidal ideation during the perinatal period include poor sleep quality,¹² domestic violence and childhood abuse,¹³

substance use,^{14,15} low socioeconomic status,¹⁶ and prior pregnancy loss.¹⁷ These risk factors are consistent across different published systematic reviews; however, to date, no umbrella review has been conducted to consolidate this evidence.

Due to the significant impact on both mothers and newborns, there has been a growing body of literature addressing suicidal ideation during the perinatal period. However, existing systematic reviews vary widely in scope, methodology, and population focus, leading to fragmented and sometimes inconsistent findings. To address these gaps and provide a higher-level synthesis of the available evidence, this study aims to perform an umbrella review that consolidates and critically analyzes existing reviews, offering a comprehensive overview of the prevalence and influencing factors of suicidal ideation during the perinatal period.

Material and methods

This umbrella review was registered in PROSPERO (registration Number: CRD42024596880) and conducted in accordance with the relevant guidelines¹⁸ and was conducted in accordance with the relevant guidelines, specifically adhering to the reporting guidelines for overviews of reviews of healthcare interventions (PRIOR).¹⁹

Search strategy and selection criteria

A comprehensive search of the literature was conducted without language restrictions in electronic databases: PubMed, Scopus, and Web of Science, from inception until November 2024. A combination of keywords and free-text terms was used, including "pregnant," "antenatal," "prenatal," "postnatal," "postpartum," "perinatal," "suicidal behav*," "suicide attempt*," "suicide," "suicidal ideation," "systematic review," and

“meta-analysis”. All citations were exported to RefWorks, where duplicates were removed. Two reviewers (SMV and RAP) independently conducted the search strategy using electronic databases and manual searches, reviewing all abstracts and titles. Full-text versions of selected articles were retrieved and reviewed to assess their eligibility. Any disagreements between the reviewers were resolved by a third reviewer (ABC). Systematic reviews (SR) with or without meta-analyses (MA) addressing the prevalence of suicidal ideation, suicide attempts, and suicide during the perinatal period, as well as articles identifying associated risk factors, were included.

Data extraction, methodological quality assessment and evidence grading

Data extraction and methodological quality assessment were independently performed by two reviewers (SMV and RAP) after reviewing the full text of the selected articles. A pre-defined data collection form was used, which included the first author's name, publication date, study time period, geographical location, number and type of primary studies included, population characteristics, and whether suicide, suicidal ideation, or suicide attempts were the primary outcome. Additionally, reported outcomes, and whether the authors performed a meta-analysis, along with global suicide prevalence and associated risk factors, were recorded.

The quality of systematic reviews (SR) was assessed using a modified version of the AMSTAR-2 tool to determine the risk of bias.²⁰ It uses a 16-item checklist covering important aspects such as study design, thorough search for information and potential bias. Each item is rated ‘yes’, ‘no’ or ‘not applicable’, and the overall score determine the review's quality. There are critical domains, such as registering the protocol before the start of the review (item 2), the adequacy of the literature search (item 4), justifying the exclusion of studies (item 7) and the risk of bias. Studies included (item 9), meta-analytic methods appropriateness (item 11), risk of bias in results interpretation (item 13), and publication bias assessment (item 15). Disagreements between reviewers were resolved by consulting a third reviewer (ABC). Each review was assigned a global quality rating of high, moderate, low, or critically low based on 16 potential deficiencies. High and moderate ratings indicated the presence of one or fewer non-critical weaknesses, while low and critically low ratings reflected one or more critical weaknesses (Appendix A).

Data synthesis

The corrected covered area (CCA) was calculated to quantify the degree of overlap between SRs. Analysis of overlap was performed by groups of SRs determined by: (a) the prevalence of suicidal ideation, suicide attempts, and suicide during the perinatal period, and (b) the suicide associated risk factors.

CCA was expressed as a percentage and calculated using the formula $(N - r) / (rc - r)$, where N represents the total number of publications included in the evidence synthesis, r is the number of rows, and c is the number of columns. A CCA below 5% indicated slight overlap, a CCA greater than

5% and up to 10% indicates moderate overlap, a CCA greater than 10% and up to 15% indicates high overlap, and a CCA greater than 15% indicates very high overlap.²¹ The findings of selected SRs were tabulated.

Results

Study selection and quality assessment

A total of 398 articles were identified. After excluding those that were non-SRs or MAs, 348 articles met the eligibility criteria. Full texts were retrieved for 22 articles, which were then reviewed. Articles on statements, recommendations, protocols, or unrelated populations were excluded, resulting in 10 articles included in the final analysis. Fig. 1 presents the PRISMA²² flow diagram of the selection process.

Characteristics of the included systematic reviews

Table 1 outlines the main characteristics of the studies included in this umbrella review, and Table 2 provides a summary of the characteristics associated with perinatal suicides from those same studies.

A total of 10 SRs were analyzed, of which 6 incorporated MAs. All reviews were published between 2018 and 2024. 4/10 (40%) SRs assessed the prevalence of suicide, suicide attempts, or suicidal ideation.^{23–26} 8/10 (80%) SRs examined risk factors associated with these conditions^{24,25,27–32}; of these, 2/8 (25%) also included information on prevalence.^{24,25}

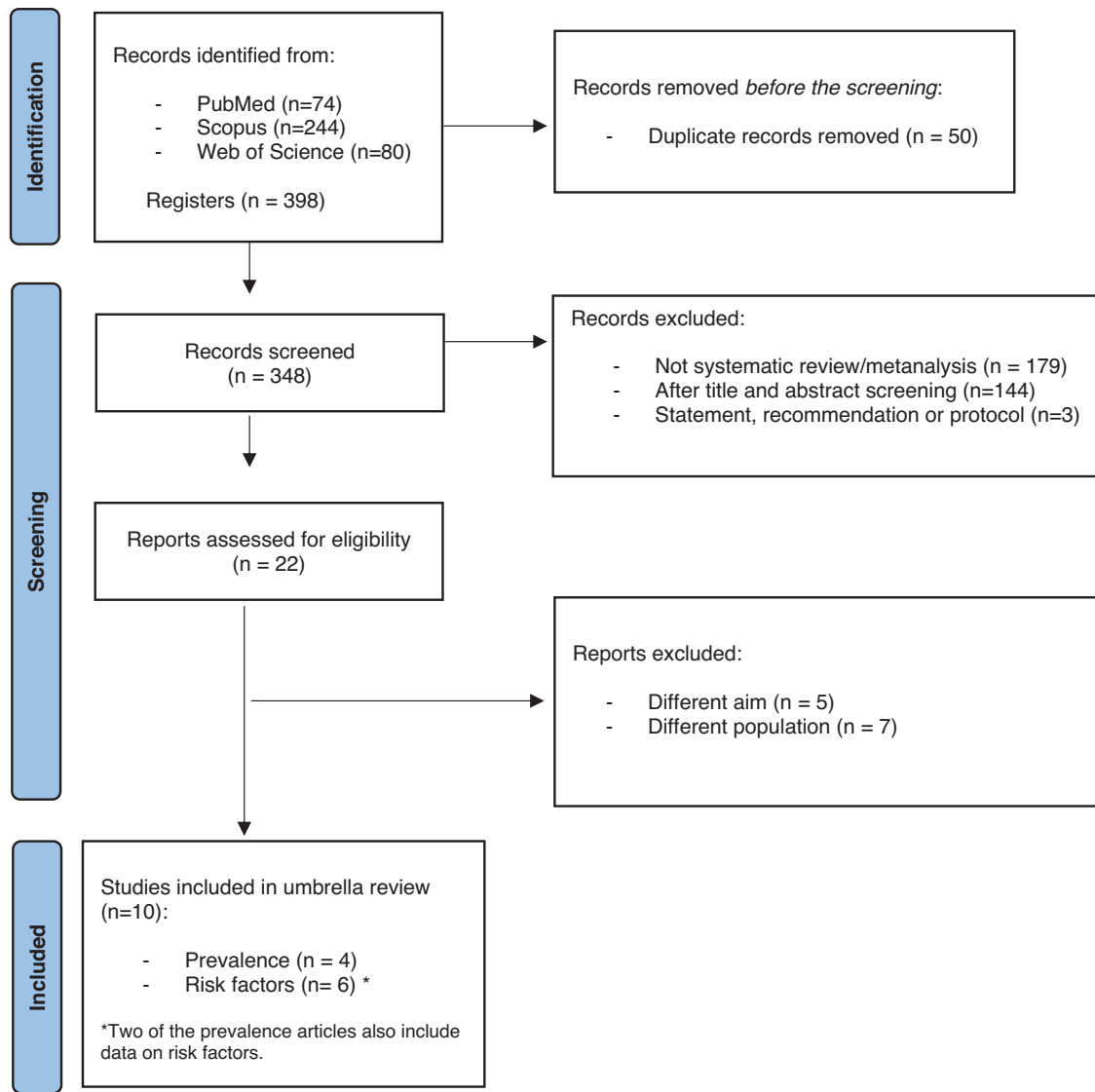
Methodological quality assessment and overlap of evidence

Using the AMSTAR 2 tool (Fig. 2), the methodological quality of the studies was assessed. One study (10%) was rated as low quality (30), while four studies (40%) were rated as critically low quality.^{27,31} Five studies (50%) received a high-quality rating.^{23,32} The most frequently identified critical weaknesses in the AMSTAR-2 criteria were related to items 2, 9, and 13, which address protocol review, risk of bias (RoB) assessment, and the consideration of RoB in interpreting results. Additionally, item 7, concerning the justification for excluded studies, was also commonly noted.

A degree of overlap was observed between SRs, with a CCA of 8.31% (Appendix B.1). The overlap in SRs addressing the prevalence of suicidal ideation, suicide attempts, and suicide during the perinatal period was 3.66% (Appendix B.2). The overlap for SRs analyzing suicide-associated risk factors was 6.40% (Appendix B.3).

Discussion

This umbrella review provides a comprehensive synthesis of the prevalence and risk factors of suicidal ideation during the perinatal period. The findings showed a varied prevalence of suicidal ideation with associated key risk factors, including poor sleep quality, intimate partner violence, and limited social support.



Source: Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

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Figure 1 Flow chart. Selected umbrella review studies on the prevalence of suicide, suicidal ideation, and associated risk factors.

This umbrella review presents several strengths, first, the inclusion of a comprehensive search strategy across multiple databases without language restrictions. Second, the adherence to PRIOR guidelines. Third, the evaluation of methodological quality using AMSTAR-2, and finally the assessment of overlap. However, certain limitations should be acknowledged. The small number of SRs and the heterogeneity among them, especially regarding definitions, methodological quality, and geographical contexts, limit the comparability of results and reduce the overall consistency of the evidence. Additionally, potential regional biases exist, as most studies originate from specific geographic areas, which may affect the global representativeness of the findings. These limitations underscore the need for future research with

more homogeneous methodologies and greater geographic diversity.

Of the 10 studies reviewed, only four provide specific data on the prevalence of suicidal ideation and suicide attempts. In these studies, the prevalence of suicidal ideation ranged from 7.2%²⁴ to 12%,²⁶ while the prevalence of suicide attempts ranged from 4% to 5.1%.^{24,26} Only one of the studies looked at the prevalence of completed suicide, which was estimated to be 4%.²⁶

This pattern shows that the percentages of suicidal ideation are higher than those of attempted or completed suicide, which may be explained by the progressive nature of the suicidal process. This process usually begins with suicidal ideation, followed by planning, communication of suicidal thoughts, and ultimately suicide attempts, which

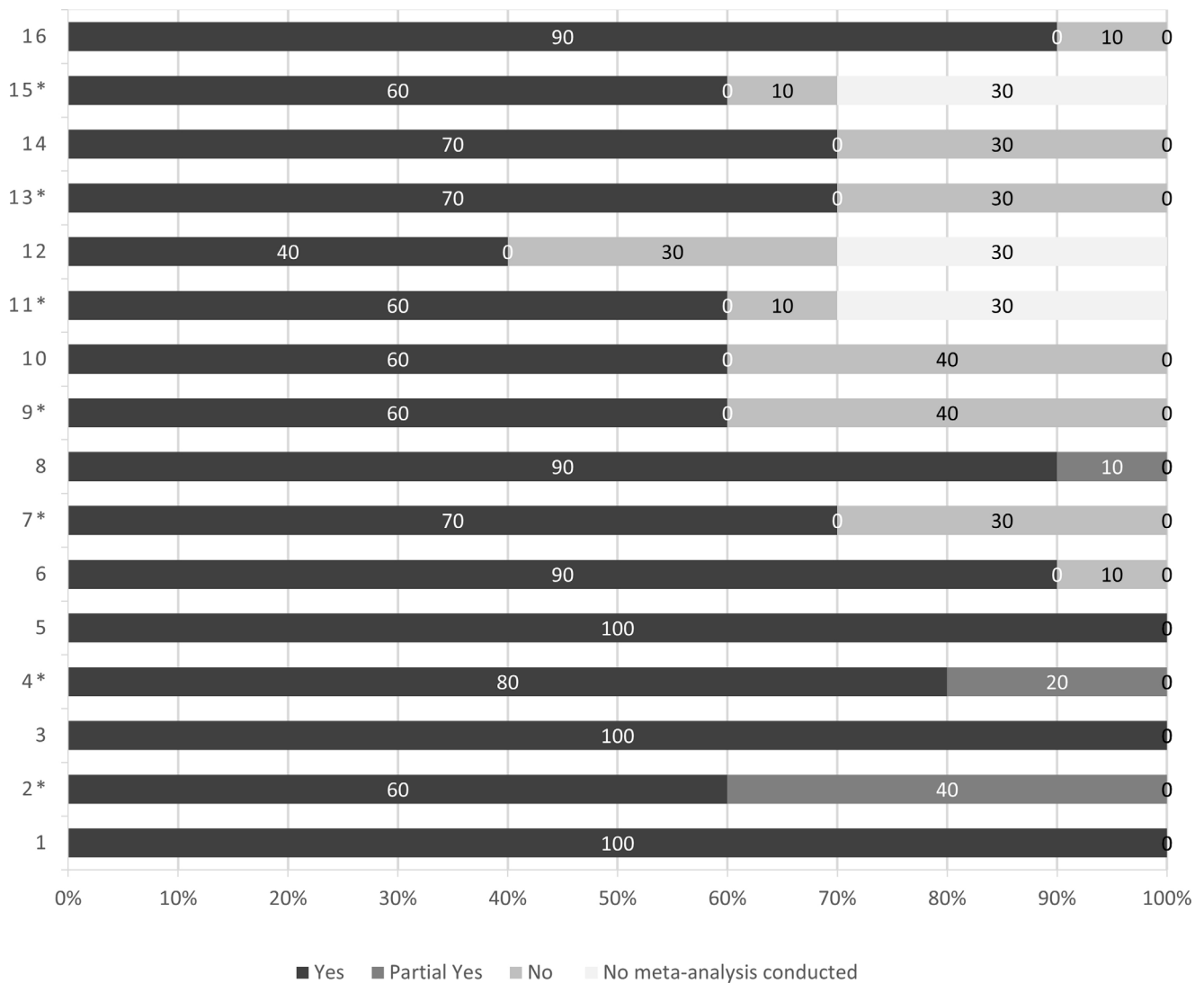


Figure 2 Methodological quality of 10 Systematic reviews included in the umbrella review of Suicidal Ideation During the Perinatal Period: Prevalence and Risk Factors. Quality assessment according to the 16 items of AMSTAR 2. *Item 1:* Research questions and inclusion criteria include PICO components; *Item 2:* Previous protocol review; *Item 3:* Explaining decision about the study designs to include in the review; *Item 4:* Adequate literature search; *Item 5:* Study selection performed in duplicate; *Item 6:* Data extraction performed in duplicate; *Item 7:* Excluded studies justification; *Item 8:* Describing included studies with sufficient detail; *Item 9:* Bias risk of individual studies included; *Item 10:* Reporting the sources of funding for the studies included in the review; *Item 11:* Appropriate meta-analysis methods; *Item 12:* Assessing the potential impact of bias risk on results; *Item 13:* Consideration of the bias risk in the interpretation of the review results; *Item 14:* Satisfactory explanation and discussing any observed heterogeneity in the review results; *Item 15:* Assessment of the presence and probable impact of publication bias; and *Item 16:* Potential sources of conflict including any funding received. *Critical items: 2, 4, 7, 9, 11, 13 and 15; Non-critical items: 1, 3, 5, 6, 8, 10, 12, 14 and 16.

may be fatal.^{33,34} This process tends to be longer in women, which increases the probability of risk factors manifesting in the intermediate stages.³³ In our review, we found that the prevalence of suicidal ideation in the perinatal period (7–12%)^{24,26} is significantly higher than the 5.28%³⁵ prevalence of suicidal ideation in the general European population. However, data from women and girls in South Asia show an even higher prevalence of 17%.³⁶ Regarding suicide attempts, our prevalence estimate (4–5.1%)^{24,26} is in line with the prevalence observed in South Asia (5%)³⁶ and globally (0.4–5.1%).³⁷ In contrast, data from the European population show a significantly lower prevalence of suicidal

behavior (2.88%).³⁵ These differences may be influenced by cultural and social differences in different contexts.

The main risk factors associated with increased suicidal ideation identified included sleep disturbance, exposure to abusive situations, domestic violence, lack of social support, and pre-existing medical conditions. Lack of adequate and restful sleep significantly increases the risk of suicidal ideation, according to our findings, and is closely related to mental health problems.³⁸ This may be because some mental health problems, such as depression, can lead to sleep problems.¹² On the other hand, lack of adequate sleep can lead to emotional lability, confusion, distorted self-

Table 1 The main characteristics of the studies included in this umbrella review.

First author	Country	Primary studies	Reported study designs included	Time of assessment	Population characteristics	TOTAL Sample size (women)
Amiri S, 2021 ²⁶	USA, Sweden, India, Japan, China, South Africa, Australia, France, UK, Nepal, Bangladesh, Italy, Canada, Argentina, Brazil, Zimbabwe, Vietnam, Ethiopia, Denmark, Qatar	37	Cross-sectional, retrospective cohort, perspective cohort	1997–2019	Postpartum women	–
Arditi-Arbel B, 2023 ³¹	Israel	51	Primarily cross-sectional studies, with some longitudinal studies.	January 1, 2015 to May 15, 2022	Perinatal Women	46
Gelabert E, 2024 ³⁰	Europa, América del Norte, Asia y África.	10	Case-control studies and cohort studies.	Until October 2021	Mothers with psychiatric disorder	–
Hasen AA, 2023 ²⁵	Ethiopia	9	Cross-sectional	During the COVID-19 pandemic	Pregnant women in Ethiopia during the COVID-19 pandemic	4.155
Karaçam Z, 2024 ²⁴	Japan, Dubai, Ethiopia, Sri Lanka, England, Nepal, Brazil, Thailand, Pakistan, Bangladesh, USA, Tanzania, South Africa, Croatia, Rwanda, and China	38	Cross-sectional and cohort studies	Between 2014 and 2021	Pregnancy, postpartum, pre-pandemic, pandemic	9.044.991
Meurk C, 2021 ²⁹	USA, Denmark, Singapore, Sweden, Finland, Israel, Canada, UK, Australia, Taiwan	18	Cohort, cross-sectional, surveillance, case-control studies	From 1996 to 11 November 2019	Perinatal women	–
O'Connor A, 2018 ²⁸	Canada, Australia, South Africa, France, Peru, United Kingdom, New York, India, Taiwan, USA, Brazil, and Sweden	15	Cross-sectional, retrospective cohort, historical cohort, prospective cohort, case-control, and two randomized controlled trials.	March 2018	Perinatal women	42.382
Palagini L, 2024 ³²	Multicentric: Peru, Ethiopia, China, Italy, Ghana, USA, and Japan.	10	Cross-sectional, longitudinal and retrospective cohort studies.	1990–April 2023	Perinatal women with insomnia or poor sleep quality	807.760

Table 1 (Continued)

First author	Country	Primary studies	Reported study designs included	Time of assessment	Population characteristics	TOTAL Sample size (women)
Reid HE, 2022 ²⁷	USA, Asia, Africa, South America, Europe, Australia.	59	Cross-sectional, cohort, case-control studies, and one used a mixed cohort and cross-sectional.	Up to June 2021	Perinatal women with or without a psychiatric diagnosis	50
Xiao M, 2022 ²³	Asia, North America, South America, Europe, Africa, Australia, Oceania	71	Cohort studies, cross-sectional studies and RCT study	August 31, 2020	Perinatal women	92.146

perception, and abnormal neural responses,^{38,39} affecting both the emotional and physical well-being of the mother, which increases the risk of suicidal ideation.⁴⁰

A personal history of abuse, both in childhood and adulthood, has been identified as a risk factor for the development of suicidal ideation in the perinatal period, a finding that was already recognized in the general population.⁴¹ Recent studies have shown that physical abuse in adulthood increases the likelihood of developing suicidal ideation by up to 27 times, while sexual abuse increases this risk by a factor of five.⁴¹ If the abuse occurred during childhood, it is associated with a 2.4 times higher risk of suicidal ideation.⁴² In particular, childhood sexual abuse has been consistently associated with suicidal behavior and is considered a significant risk factor for attempted suicide.⁴³

Domestic violence (IPV) also increases the likelihood of developing suicidal ideation in the perinatal period. This finding is consistent with previous studies,⁴⁴ which estimate that 42.9% of women who experience IPV have suicidal ideation.⁴⁵ In turn, IPV has been considered by some authors to be a risk factor for depression, which increases the probability of suicidal ideation.⁴⁶ These findings are consistent with those of a recent MA of women in the general population who had experienced IPV.⁴⁷

Similar to domestic violence, lack of social support has been identified as a factor that facilitates the development of suicidal ideation. Lack of social support is associated with higher rates of anxiety and depression, both prenatally and postnatally.^{48,49} In contrast, high levels of social support are associated with better overall mental health, including significant reductions in depressive symptoms.⁵⁰ Some studies even suggest that strong social support may play a protective role against the development of mental illness during pregnancy.⁵¹⁻⁵³

Not only social factors play an important role in suicidal ideation, but also the presence of pre-existing medical conditions, particularly those related to mental health problems, predisposes to suicide.^{44,54} This is consistent with previous research showing that disorders such as anxiety, stress, depression, and even post-traumatic stress disorder are strongly associated with increased risk.^{46,55} Overall,

women with mental health problems during pregnancy are three times more likely to develop suicidal ideation than women without mental health problems during pregnancy.⁵⁶ Although an increased risk has also been found in women with physical illnesses such as cancer or epilepsy, this is much lower than the risk associated with psychiatric disorders.⁵⁷

Suicidal ideation and postpartum depression affect a mother's ability to deal with the stress of parenting, interfere with her interactions with her baby, and affect the mother-child bond.^{58,59} This bond has a major impact on the mother's self-concept and emotional well-being, and its deterioration may affect the mother's ability to respond appropriately to her baby's needs, leading to neglectful or inappropriate behavior.⁵⁹

It is imperative to recognize the significance of the perinatal period in maternal well-being, particularly the potential for the mitigation of risk factors that contribute to suicidal ideation. To support mother's mental health during this time, health professionals are essential in recognizing and addressing these factors. The identification of modifiable risk factors such as poor sleep quality, intimate partner violence, or limited social support offers concrete opportunities to improve clinical practice. These findings underscore the importance of routinely integrating psychosocial screening into perinatal care. Healthcare professionals should be trained to assess sleep disturbances, detect signs of interpersonal violence and evaluate the presence or absence of support networks during both prenatal and postnatal visits. Early identification can facilitate timely referral to mental health services and allow for the implementation of more targeted interventions.

Implementing focused public health policies that prioritize maternal mental health screening and intervention is therefore vital. The adoption of these strategies within a multidisciplinary care model can strengthen health systems' ability to prevent mental health disorders during this vulnerable period. Future studies should concentrate on clarifying the mechanisms underlying these risk factors and evaluate effective preventive interventions. Clinically, both mothers

Table 2 The main characteristics of the studies on Perinatal Suicide included in this umbrella review.

First author	Meta-analysis (Y/N)	Prevalence of suicidality (Y/N)	Prevalence of suicidal ideation (Y/N)	Prevalence of attempt (Y/N)	Outcomes	Associated factors
Amiri S, 2021 ²⁶	Y	4%	12% (95% CI: 11–14)	4% (95% CI: 1–7)	Prevalence of postpartum suicidal ideation, suicide attempts, and suicide mortality	Depression
Arditi-Arbel B, 2023 ³¹	N	N	N	N	Risk factors for suicidal ideation and behavior	- Sleep disturbances - Psychopathology (depression, anxiety, etc.) - Low social support - Medical conditions (such as HIV) - Adverse childhood experiences - Intimate partner violence
Gelabert E, 2024 ³⁰	Y	N	N	N	Risk factors for suicidal attempts	- Sociodemographic: (Marital status, education level, age) - Clinical (Mood disorders, history of suicidal behavior) - Obstetric and neonatal (cesarean delivery, low birth weight, perinatal loss).
Hasen AA, 2023 ²⁵	Y	N	11% (95% CI: 0.09, 0.13).	N	Prevalence of suicidal ideation	- Depression - Intimate partner violence - Age (≥ 30 years) - Stress - Poor sleep quality - Substances
Karaçam Z, 2024 ²⁴	Y	5.1% (95% CI, 0.01–1.53)	7.8% (95% CI: 0.06–0.11)	1% (95% CI: 0.00–0.07)	Prevalence and frequency of suicidal behavior, frequency of suicidal behavior in the pre- and post-COVID-19	- Mental health problems - Insomnia - Various forms of violence - Younger age - Being unmarried - Low education and income - Unplanned pregnancy - History of abortion, trauma, and lack of social support - The presence of VIH

Table 2 (Continued)

First author	Meta-analysis (Y/N)	Prevalence of suicidality (Y/N)	Prevalence of suicidal ideation (Y/N)	Prevalence of attempt (Y/N)	Outcomes	Associated factors
Meurk C, 2021 ²⁹	N	N	N	N	Risk factors for suicidal behavior	- Mental health - Pregnancy and Health - Interpersonal relationships and intimate partner violence - Health service utilization
O'Connor A, 2018 ²⁸	N	N	N	N	Risk factors for suicidal ideation	- Low socioeconomic status - Prior mental health disorders - Intimate partner violence - Smoking, history of abuse - Migraine headaches.
Palagini L, 2024 ³²	Y	N	N	N	Risk factors for suicidal ideation	- Insomnia - Poor sleep quality
Reid HE, 2022 ²⁷	N	N	N	N	Risk factors for suicidal ideation	- Negative life events (childhood maltreatment only or adverse childhood experiences; maltreatment at any time in life (childhood and/or adulthood): adult maltreatment only or intimate partner violence (IPV); and stress (e.g. stress during pregnancy or postpartum; exposure to traumatic life events; post-traumatic stress symptoms and first-degree relative suicide). - Social factors (risk factors related to mother–child bonding and mother-child interactions; social support and loneliness) - Cognitive factors (guilt, shame and worthlessness – personality and individual differences (self-esteem/self-efficacy)
Xiao M, 2022 ²³	Y	N	10% in pregnancy and 7% in the postpartum.	N	Prevalence of suicidal ideation	-

Abbreviations: Y: yes, N: no, CI: confident interval.

and their newborns can benefit from routine perinatal care that incorporates comprehensive mental health support.

Conclusion

The perinatal period is a critical window for maternal mental health, with suicidal ideation being influenced by modifiable risk factors. Health professionals play a vital role in identifying and addressing these factors in order to support the mental well-being of mothers during this time.

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Conflict of interest

The authors declare that they have no conflict of interest.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.semern.2025.102567](https://doi.org/10.1016/j.semern.2025.102567).

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